Denny Schaedig

Machine Learning Engineer

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SUMMARY

ML engineer who designs, trains, and ships ML systems end to end, from data pipelines and distributed training to model packaging and serving. Published researcher (Neurophotonics, 2025) with hands-on ownership of HIPAA-compliant cloud HPC and a self-hosted GPU cluster. Experienced across generative models, deep learning, LLM agents, and pre-registered, ablation-controlled evaluation.

SKILLS

Languages: Python, SQL, TypeScript / JavaScript, Dart, Bash
Machine Learning & Deep Learning: PyTorch, TensorFlow / Keras, Hugging Face Transformers, scikit-learn, OpenCV; generative models (GANs, diffusion), transformers, transfer learning, computer vision, multimodal fusion
LLMs & Agents: Agentic systems, LLM evaluation & hallucination detection, RAG (pgvector, sentence-transformers), local inference (llama.cpp), Anthropic & OpenAI APIs
MLOps & Infrastructure: AWS (EC2, Lambda, S3, EFS, Glacier), Docker, Ray, CUDA, Weights & Biases, PostgreSQL, Prometheus, GitHub Actions, pytest
Serving & Distribution: FastAPI inference APIs, on-device inference (TensorFlow Lite), Hugging Face Hub, PyPI packaging (hrfunc, pymaxim), Flutter (iOS & Android)
Scientific & Biomedical Data: fMRI, fNIRS, physiological signals; MNE / MNE-NIRS, Nilearn; statistical modeling (SciPy, statsmodels, pandas); protected health data (HIPAA); experimental design (pre-registration, ablations)

PROJECTS

- **HRfunc** - Open-source Python package for hemodynamic signal deconvolution with a live peer-to-peer HRF-sharing database; published in Neurophotonics (2025) and demoed in an invited NIRx vendor webinar. hrfunc.org
- **Maxim** - Open-source LLM agent harness (PyPI: pymaxim, Apache-2.0; ~13K downloads) with persistent cross-session memory, learned drives, and RAG; every release ships pre-registered, ablation-controlled experiments, nulls included (e.g., a taught want transferred between agents: 0.84 vs. 0.22 isolated, n = 50 per arm). pymaxim.bio
- **snowGAN / Rocky Mountain Snowpack** - Hand-collected 4,040-sample CV dataset (7 snowpits, two winters; 40K+ Hugging Face downloads; DOI 10.57967/hf/6143) and pretrained GANs, a transfer-learning backbone for avalanche-risk prediction (in development). rmdig.ai

EXPERIENCE

Laboratory for Child Brain Development — Washington University in St. Louis — St. Louis, MO
NeuroImaging Engineer (Machine Learning & Software Engineering) — *Dec 2022 – Present*

- Architected a HIPAA-compliant AWS HPC environment (EC2, EFS, S3, Glacier) ~32% under projected cost and secured an additional 24% university-wide cloud discount.
- Designed and shipped **HRfunc**, an open-source, pip-installable Python package for signal deconvolution; published in Neurophotonics (2025), presented in an invited NIRx vendor webinar, and maintained across 22 public releases (~18K PyPI downloads).
- Trained **NIRSgen**, a diffusion model trained from scratch on biomedical time-series for data augmentation; validated via transfer learning, lifting downstream classification accuracy to 92% (vs. 76% randomly reinitialized).
- Built **BOLDnet**, a TensorFlow framework with custom spatial-attention layers and 4D volumetric data loaders for rapid model prototyping on standardized datasets.
- Developing **synchronAI**, a multimodal deep-learning AI fusing video, audio, and time-series signals for automated inference (0.72 AUC) (github.com/dennys246/synchronAI).

Rocky Mountain Digerati, LLC — Summit County, CO
Founder & ML Engineer (independent applied-ML lab) — *Nov 2025 – Present*

- Built an LLM analyst agent for a healthcare-finance contract client, including a hallucination detection and removal layer, running on open-weight models as small as 16B parameters (in client beta).
- Run a self-hosted 3-GPU Ray cluster (mixed CUDA and Apple Silicon backends, Tailscale mesh) for local training, evaluation, and model serving.

- Collected and curated the Rocky Mountain Snowpack dataset: 4,040 image samples from 7 snowpits across two winters, published on Hugging Face (DOI 10.57967/hf/6143; 40K+ downloads).
- Generatively pre-trained snowGAN (WGAN-GP) on snowpack imagery and released the weights on Hugging Face as a transfer-learning backbone for avalanche-risk modeling (in development).
- Built AvAI, a backcountry check-out safety app for iOS and Android (Flutter), launching fall 2026; prototyped on-device TensorFlow Lite inference in test builds.
- Built AvServ, AvAI's Prometheus-monitored rescue-initiation backend, which alerts search and rescue when a user misses their expected check-in.

The Pain Lab — University of Colorado Boulder Boulder, CO
Software / ML Engineer *Jul 2020 – Feb 2023*

- Trained an RNN to predict pain levels from wearable-sensor and questionnaire data (79% accuracy vs. 63% majority-class baseline).
- Built ETL pipelines moving and preprocessing data from remote servers and devices into a relational SQL database, time-aligning multiple physiological streams for ML training.

Intermountain Neuroimaging Consortium — University of Colorado Boulder Boulder, CO
Software Engineer *Jul 2017 – Feb 2022*

- Built ETL pipelines and automation for MRI scheduling, study enrollment, and training-compliance, improving recruitment throughput and data availability.

PUBLICATIONS

Graff, K., Zhang, Y., **Schaedig, D.**, Melhem, N., & Perlman, S. B. *Stability of the functional connectome across childhood: A five-year dense-sampling case study with multi-subject validation.* NeuroImage (2026). doi.org/10.1016/j.neuroimage.2026.122246

Rocha-Hidalgo, J., Nguyen, H., Reis, D., **Schaedig, D.**, Thompson, K. I., Perlman, S. B., & Pérez-Edgar, K. E. *Parent–child neural dynamics during a stressful task: Probing associations with temperament and anxiety.* Clinical Psychological Science (2026). doi.org/10.1177/21677026251396786

Thompson, K. I., Zhang, Y., **Schaedig, D.**, Bankston, A., Parnerkar, A., Jeyaram, S., Melhem, N., & Perlman, S. B. *Identifiable regional and neural network abnormalities in early childhood anxiety.* JCPP Advances (2026). doi.org/10.1002/jcv2.70158

Schaedig, D., Schumer, M., Mata-Centeno, B., Pollonini, L., Pérez-Edgar, K., Melhem, N., & Perlman, S. B. *HRfunc: A tool for modeling hemodynamic response variability in fNIRS.* Neurophotonics (2025). doi.org/10.1117/1.NPh.13.S1.S17801

Alvarado, C., Panlilio, C. C., Pérez-Edgar, K., Thompson, K. I., **Schaedig, D.**, Melhem, N., & Perlman, S. B. *Neural synchrony in parent-child dyads.* bioRxiv (2025). doi.org/10.1101/2025.07.02.662817

EDUCATION

M.S., Artificial Intelligence & Machine Learning - Colorado State University Global 2025
B.A., Neuroscience - University of Colorado Boulder 2019

CERTIFICATES & AWARDS

- Stanford Online - Algorithms: Divide & Conquer / Sorting & Searching; Graph Search, Shortest Paths & Data Structures.